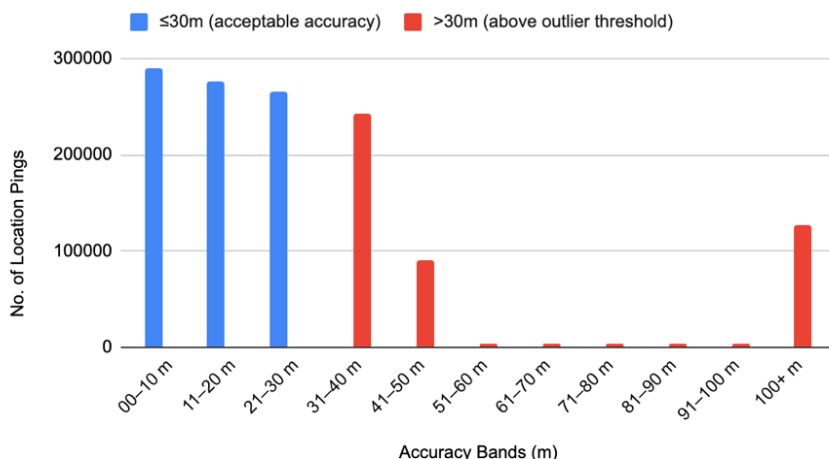


Part 1 – Data Presentation (descriptive Statistics & Key Visuals)

This section summarises key descriptive patterns and insights. Figures 1A–1C show single-variable distributions, while Figures 1D–1F highlight relationships between variables.

Figure 1A – Location Accuracy Distribution (Yavne branch)

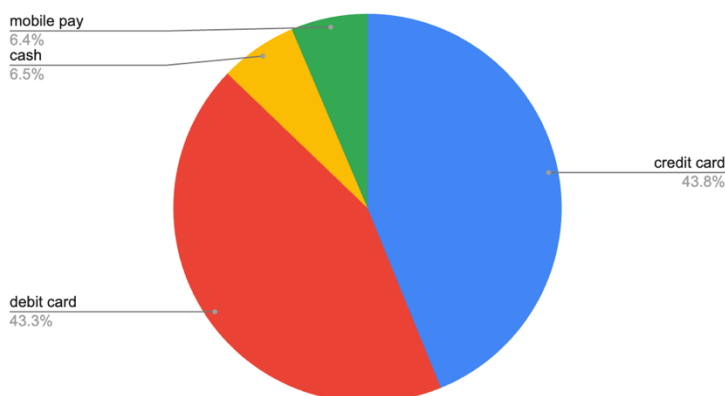
Figure 1.A Distribution of Location Accuracy by Device (Yavne Branch)



Approximately 74.2% of pings are within ≤ 30 m accuracy, with a small right-tail including a clear 100m+ outlier group. **Implication:** treat >30 m as lower-confidence signals (filter or flag) to avoid biasing zone/behaviour analysis (Query 1.1A).

Figure 1B – Payment Method Distribution

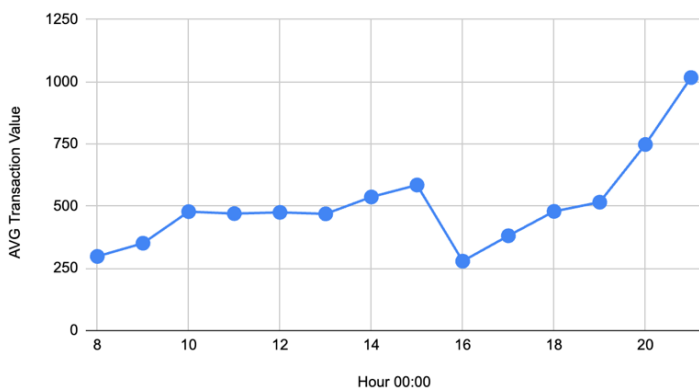
Figure 1.B Payment Method Distribution % (Yavne Branch)



Payments are overwhelmingly cashless (~93.5% digital), with cash a small minority (~6.45%). **Implication:** checkout operations should optimise digital payment flow while maintaining minimal cash-handling capacity (Query 1.1B).

Figure 1C – Average Transaction Value by Hour

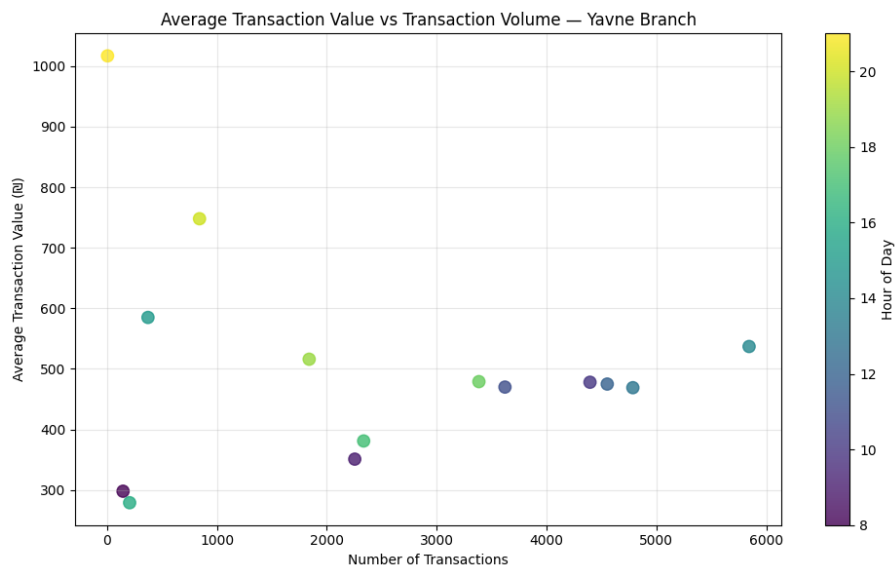
Figure 1.C AVG Transaction Value by Hour 00:00 (Yavne Branch)



Average basket value is relatively stable during core hours (~10:00–15:00) and increases later in the day; extreme late-hour values likely reflect low transaction counts. **Implication:** interpret “high averages” alongside volume to avoid over-reading late-hour noise (Query 1.1C).

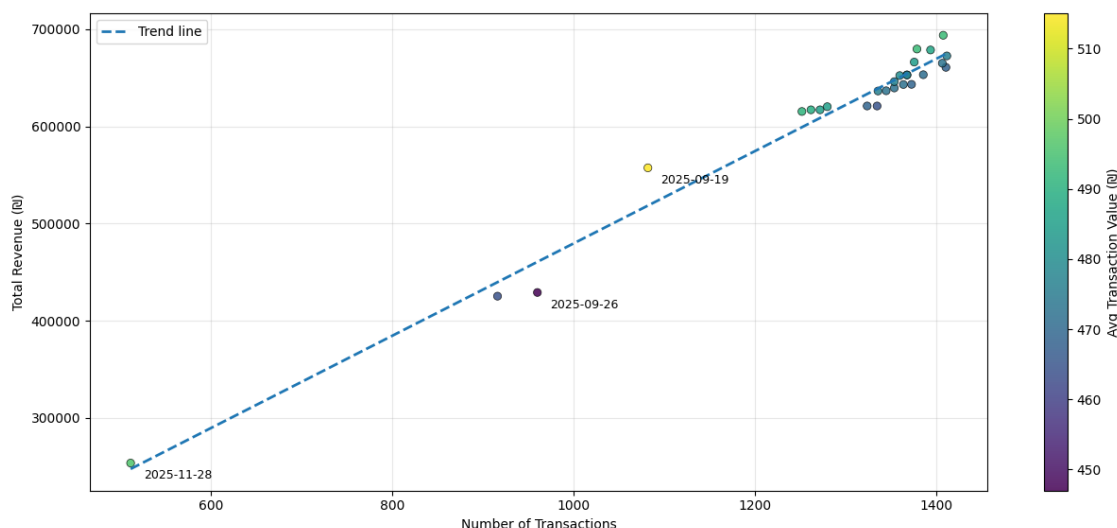
Figure 1D – Avg Transaction Value vs Transaction Volume

High-volume periods align with



moderate basket values, while unusually high averages appear mainly during low-volume hours. **Implication:** performance decisions (pricing/ staffing/ promos) should be volume-weighted, not based on isolated high averages (*Query 1.1D*).

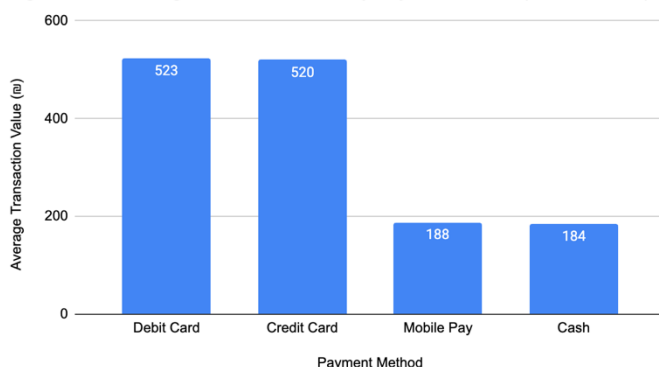
Figure 1E – Revenue vs Transactions (Weekly)



Weekly revenue tracks transaction volume closely, indicating revenue is primarily volume-driven rather than basket-driven. **Implication:** throughput and visit frequency are the key levers; deviations may reflect holidays/partial operating periods (*Query 1.1E*).

Figure 1F – Avg Txn Value by Payment Method

Figure 1.F — Average Transaction Value by Payment Method (Yavne Branch)



Digital payments produce substantially higher basket sizes (debit ≈₪523, credit ≈₪520, mobile pay ≈₪188) and cash (≈₪184). **Implication:** higher-value purchasing is strongly associated with card usage; encouraging card/digital adoption supports higher revenue per transaction (*Query 1.1F*).

Part 2 – Metrics

Q1. Customer segmentation (behavioural)

Customer devices were classified into behavioural segments using in-store geolocation presence and payment evidence (*Queries 2.1.1–2.1.6*). Across the period, 1,037 distinct customer devices were observed in-store.

Final segments (*Query 2.1.6*):

- **Repeat customers: 618 devices** (loyal core; weekly presence + repeated paid visits).
- **One-time customers: 143 devices** (observed in one visit week; low retention).
- **Not-paying visitors: 276 devices** (in-store presence with no app-payment linkage; possible friction/abandonment).
- **No-phone customers: 3,596 transactions** where `customer_id` IS NULL (non-app payments → visibility gap, not necessarily weak demand) (*Queries 2.1.2, 2.1.3, 2.1.6*).

Q2. Employee mapping (internal segmentation)

An employee roster of 43 distinct devices was identified and assigned to roles using area exposure, dominant location, and rule-based inference (*Queries 2.2.2–2.2.6*), to support staffing and shift-level analysis in later sections.

Employee list by role (43 devices; *Query 2.2.6*):

Manager (1): man_001
Cashiers (15): cas_001 – cas_015
Butchers (4): but_001 – but_004
General workers (10): gen_001 – gen_010
Security (4): sec_001 – sec_004.
Delivery/suppliers (8): del_001 – del_008
Senior general worker (1): sen_001

BONUS: role-free inference logic (no `geolocation.role`) implemented (*Query 2.2.7*), which uses only broadcast behaviour (dominant area + activity regularity) and payment evidence from `log_sales` (`paid_with_app`) to assign each device to a segment. The segments we inferred mostly matched the roles in the dataset, which suggests the method works reliably.

Q3. Outliers & App Performance

(a) Accuracy outliers (`accuracy_m` > 30m): 415,580 / 1,308,624 pings (31.76%) are classified as low-precision. These outliers are not evenly spread — they are concentrated in PARKING (54.79%, $p_{99} \approx 114m$) and among security devices (68.65%, $p_{99} \approx 149m$). Implication: they're

acceptable for broad “presence” signals, but should be treated as low-confidence for any fine movement / path analysis. (*Query 2.3.1*)

(b) Broadcast performance vs 40% benchmark: Using the sessionised PARKING test, the app shows a weighted capture rate of 0.5250 across 26,790 sessions. Only 6.40% of sessions fall below the 40% benchmark, which indicates the broadcasting performance generally meets the requirement overall. (*Query 2.3.2*)

(c) **Other outliers:** We identified 38 device outliers with unusually high ping volumes (mostly employees + a few customers). These devices can bias averages and per-device KPIs, so we exclude or handle them separately in customer analyses. (*Query 2.3.3*)

Q4. Supplier Deliveries (Delivery Guys)

Using role=delivery_guy plus delivery signature (05:30–06:30, WAREHOUSE/ ENTRANCE/ PARKING), we identified 8 supplier devices (del_001–del_008) with repeat early-morning appearances (*Query 2.4.1*). Behaviour matches recurring early-morning deliveries (9–16 delivery dates; 95–171 pings).

BONUS: The same 8 supplier devices were recovered without using the role field, using only timing + location pattern + repeat-threshold rules (*Query 2.4.2*).

Missed deliveries (no supplier arrivals recorded in the expected window): 2025-06-02, 2025-06-26, 2025-08-07, 2025-10-02, 2025-11-27 (*Query 2.4.3*).

Q5. Top customers (≈100voucher candidates)

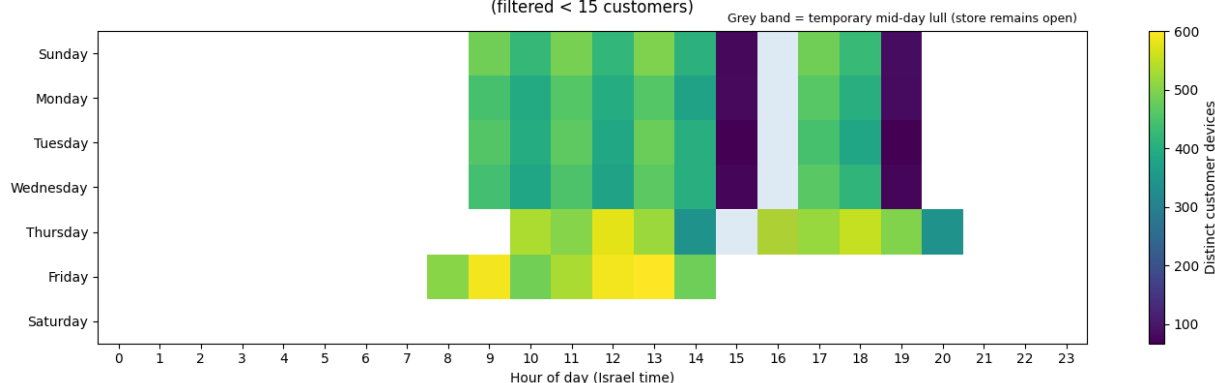
Top 5 customers by visit frequency (*Query 2.5.1*): **cus_003** = 146 visit days, **cus_011** = 139, **cus_021** = 133, **cus_045** = 129, **cus_078** = 125.

Top 5 customers by cumulative spend (*Query 2.5.2*): **rep_207** = ₪51,527.53 (98 transactions), **rep_182** = ₪51,237.57 (100 transactions), **rep_076** = ₪50,578.71 (93 transactions), **rep_171** = ₪49,391.87 (92 transactions), **rep_217** = ₪49,181.26 (90 transactions).

Q6. True operating hours (observed presence)

Using customer geolocation presence, we inferred the supermarket’s “true” operating hours from observed in-store activity (customers only; employees excluded). To avoid treating sporadic/edge pings as real trading activity, hours with fewer than 15 distinct customer devices were treated as noise and excluded from the operating-hours signal (*Queries 2.6.1–2.6.2*).

Figure 2.6 — Customer Presence by Hour and Weekday
(filtered < 15 customers)



<u>Weekday</u>	<u>Opening Hour</u>	<u>Closing Hour</u>	<u>Note</u>
Sunday	09:00	20:00	Full workday
Monday	09:00	20:00	Full workday
Tuesday	09:00	20:00	Full workday
Wednesday	09:00	20:00	Full workday
Thursday	10:00	21:00	Later open + later close
Friday	08:00	15:00	Erev Shabbat (short day)
Saturday	—	—	Shabbat (closed)

Part 3 – Business Recommendations (Customer Exp. & Operations)

Q1. Recommended quiet shopping windows (for Shani)

Customer presence is lowest right after opening, during a mid-afternoon lull, and near closing (Figure 2.2.6; *Queries 2.6.1–2.6.2*). **Recommended low-wait windows: Sun–Wed 09:00–10:00, Thu 10:00–11:00, Fri 08:00–09:00, plus 15:00–16:00 (Sun–Thu).** These windows are consistently “quiet” because demand clusters around work/school routines, while midday trips tend to be more top-up oriented. Friday demand also compresses earlier due to Shabbat timing, making the morning the safest low-wait window.

Q2. Peak vs calm patterns

Checkout demand is highly predictable (*Query 3A.2.1*): strongest pressure occurs **Thu ≈11:00–13:00 and ≈19:00**, and **Fri ≈13:00–14:00**, with the maximum spike at **Fri 14:00** (2,487 transactions) and the highest paying-customer hour at **Fri 13:00** (399 customers) (*Query 3A.2.1*). Demand drops sharply late and shows a repeatable 15:00–17:00 lull, which creates a clear operational rhythm: staff for short peak bursts, protect the Thursday/Friday windows, and use the lull for demand-shifting. In practice, queue risk is concentrated in a small number of “known” hours rather than spread evenly across the week.

Overall, shopping behaviour is routine-driven: weekday peaks align with work/school schedules, while Thu–Fri demand reflects ‘stock up’ trips ahead of Shabbat/closures (*Query 3A.2.1*).

Q3. Staffing, self-checkout, and sensitivity

(a) Staffing logic (cashiers per hour): Staffing was estimated from transactions per hour as the checkout workload metric, using transactions as a proxy for checkout customers ($\approx$ one customer per transaction), and applying a capacity assumption of ~ 25 transactions per cashier per hour (*Queries 3A.2.1; 3A.3.1*). This assumption can be validated quickly with a short POS/observation check during peak periods (e.g., measure throughput per cashier and whether queues build). Once validated, it provides a repeatable rule-of-thumb for scheduling.

Recommended cashier coverage by peak window: (*Queries 3A.2.1; 3A.3.1*)

- **Thu peaks (11:00–13:00, 17:00, 19:00):** ~ 3 cashiers
- **Fri late morning–early afternoon:** 4–5 cashiers (max at Fri 14:00)
- **Most other open hours:** 1–2 cashiers

(b) When to prioritise self-checkout: Self-checkout should be concentrated where queue risk is highest: **Thu 11:00–14:00, 17:00–20:00; Fri 09:00–14:00** (*Query 3A.3.3*). Outside these windows, staffed lanes are usually sufficient and self-checkout adds limited operational value (*Query 3A.3.3*). A practical approach is assisted self-checkout in peaks (to avoid small frictions becoming bottlenecks), then scaling it down during the mid-afternoon lull.

(c) What would change the recommendations: Chagim/pre-Chagim timing, promotions/discounts, payday/month-end effects, extreme weather, register/system downtime, and staffing shortages can shift peaks or reduce capacity and should trigger temporary re-planning. The key risk is not only higher volume, but also slower processing (larger baskets, more exceptions) that reduces effective checkout capacity.

Q4. No-activity dates and spike dates (planning context)

Closures (no activity): every Saturday (2025-06-07 to 2025-11-29), plus 2025-06-02, 2025-09-23–2025-09-24, 2025-10-02, 2025-10-07, 2025-10-14 (*Query 3A.4.1*) — consistent with major Chagim closures and should be treated as expected calendar effects.

Demand spikes (≥ 95 th percentile): 2025-06-05, 2025-06-20, 2025-07-03, 2025-07-17, 2025-08-29, 2025-09-04, 2025-09-11, 2025-11-13, 2025-11-27 (*Query 3A.4.2*) — consistent with Erev-Chag “stock-up” behaviour. These dates are suitable for proactive actions (surge lanes, queue management support, fewer break overlaps).

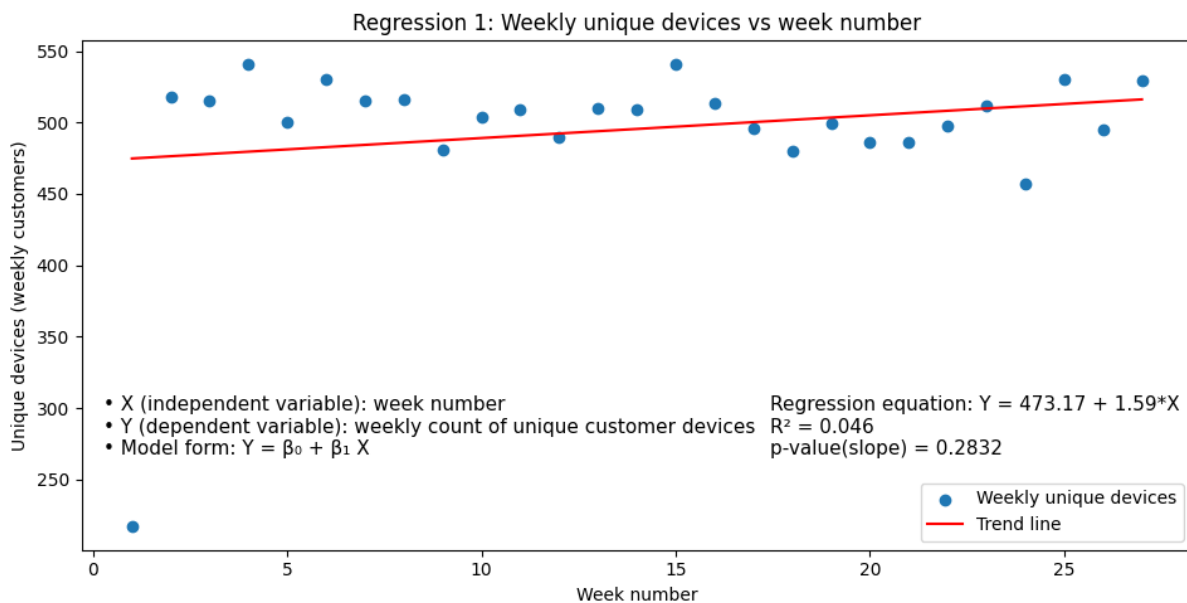
Q5. Balancing experience and profitability (practical tools)

Tools & service scenarios to improve the shopping experience: express lane (≤ 10 items), assisted self-checkout in peak windows, Click-&-Collect / pre-picked baskets, quiet-hour nudges, and faster payment flow (tap-to-pay priority, digital receipts, auto-applied coupons). The goal is to protect time-pressed shoppers during peaks without permanently inflating labour costs.

“Calm Hours” program: classify each hour as peak/shoulder/calm using presence + checkout evidence (Figure 2.6; *Queries 2.6.1–2.6.2; Query 3A.2.1*), then activate small incentives only in calm windows (e.g., **15:00–16:00 Sun–Thu**) to shift discretionary trips. Examples: time-limited loyalty points, a small discount on a defined basket, or priority Click-&-Collect pickup—paired with “best time to shop” messaging for customers who prefer low congestion.

Part 4 Regression Models: Visits & Spend

Regression 1 — Weekly unique customer devices vs. week number (*Query 4.1*)



Model: Unique customer devices (weekly) = $\beta_0 + \beta_1 \times (\text{week number})$ (*Query 2.4.1*)

H0 (null): $\beta_1 = 0 \rightarrow$ week number does not predict weekly unique devices (no linear trend).

H1 (alt.): $\beta_1 \neq 0 \rightarrow$ week number is a statistically significant predictor (a trend exists).

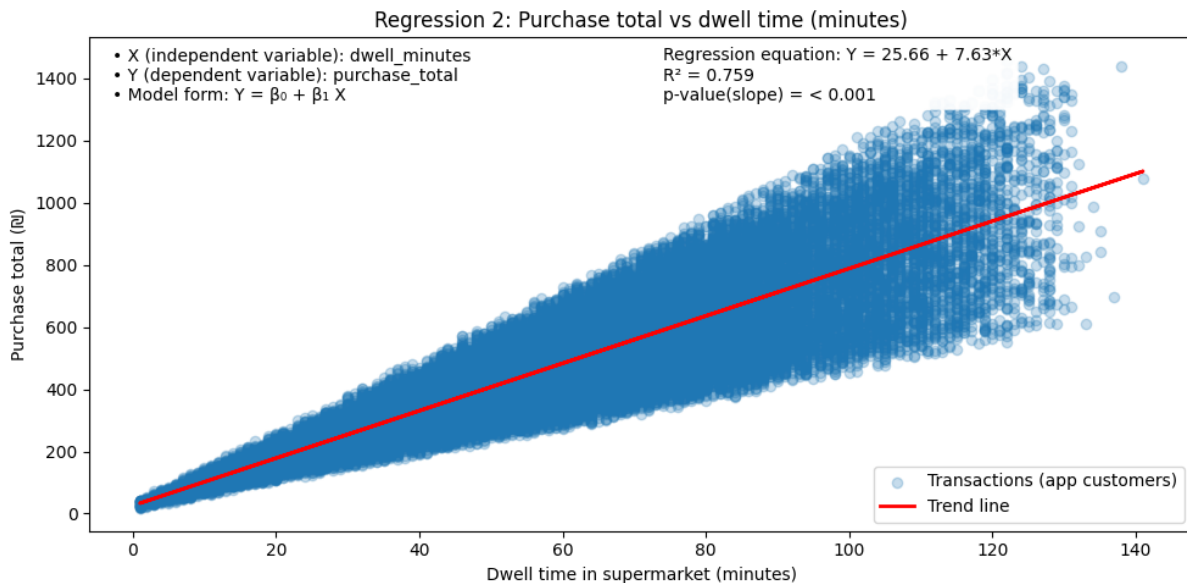
Decision: $p = 0.283 \rightarrow$ fail to reject H0; no statistically significant linear trend in weekly unique devices.

Result & interpretation: Fit is weak ($R^2 = 0.046$) (*Query 4.1*). Although the slope is slightly positive (+1.59 devices/week), weekly customer traffic is stable over time and the week-to-week variation is better explained by calendar noise/seasonality than by sustained growth.

Business recommendation: Don't assume organic growth. Profit growth should come from planned demand drivers (quiet-hour incentives, loyalty reactivation, and calendar-based campaigns), rather than expecting a steady week-by-week lift.

What data would improve this model: Add a Chag/Erev-Chag flag, a promotion/discount calendar, and external demand shocks (weather / local events / competitor disruption) to separate “true demand” from calendar-driven volatility.

Regression 2 — Purchase total vs. dwell time (minutes) (Query 4.2)



Model: Purchase total (₪) = $\beta_0 + \beta_1 \times (\text{dwell minutes})$ (Query 4.2)

Sample: App customers only (non-NULL dwell_minutes); rows with missing dwell_minutes were excluded. (Query 4.2)

H0 (null): $\beta_1 = 0 \rightarrow$ dwell time does not predict purchase total (no linear trend).

H1 (alt.): $\beta_1 \neq 0 \rightarrow$ dwell time is a statistically significant predictor (relationship exists).

Decision: $p < 0.001 \rightarrow$ reject H0; dwell time is a statistically significant predictor of purchase total (positive linear relationship).

Result & interpretation: Fit is strong ($R^2 = 0.759$) (Query 4.2). Each additional minute in-store is associated with ~₪7.63 higher spend, suggesting longer visits generally translate into larger baskets rather than random variation.

Business recommendation: Increase profit by lifting basket size per trip: bundles/meal-deals, in-aisle upsell near fresh/bakery/meat, and targeted offers for app users while they are in-store. Operationally, also ensure dwell time reflects shopping, not queue/wait friction.

What data would improve this model: Add item count / basket mix (to distinguish “big basket” from “quick top-up”), a queue / cashier-availability-by-hour measure (so dwell isn’t inflated by congestion), and customer segment / loyalty tier to test whether the dwell→spend relationship differs across customer types.

Part 5 – Looker Studio Dashboard (Charts & Insights)

Dashboard Overview & Filters

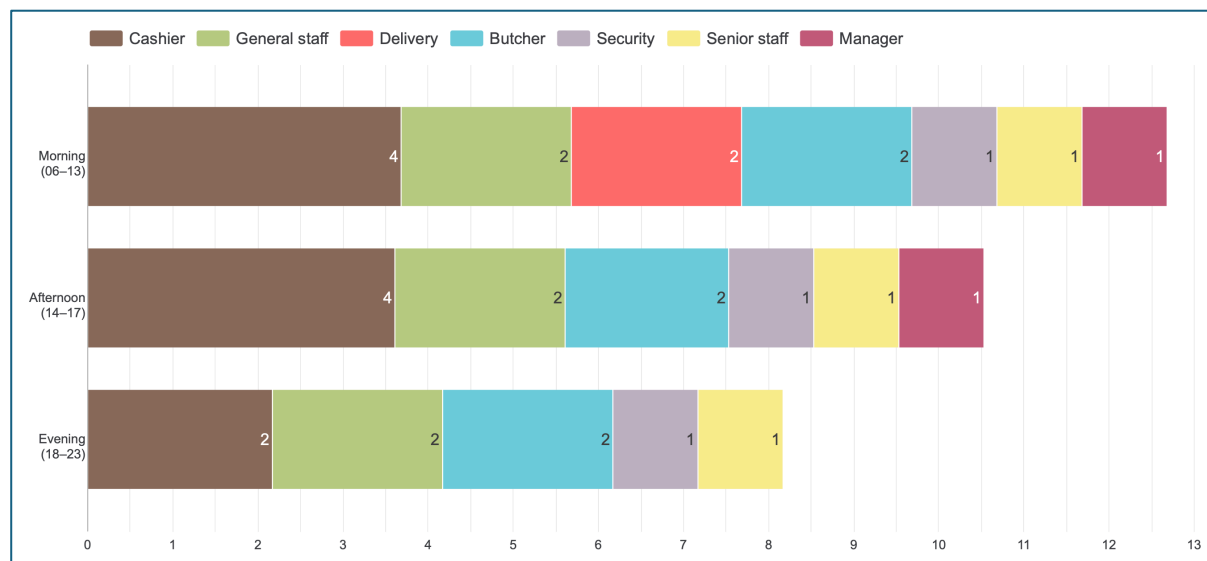
The Looker Studio dashboard consolidates customer behaviour and staffing insights for the supermarket and is intended for management monitoring and decision-making. All visuals support an interactive date range control (“Select date range”) so trends and distributions can be reviewed for any chosen period.

Figure 5.1 Activity Heatmap (Uniq. Devices) - Hour×Day of Week (Query 5.1)

hour	1 Sunday	2 Monday	3 Tuesday	4 Wednesday	5 Thursday	6 Friday
8						1,775
9	927	834	805	822		3,611
10	748	671	662	660	2,230	2,544
11	1,000	905	871	886	2,266	2,385
12	770	686	630	633	3,055	3,559
13	1,034	887	933	937	2,281	4,373
14	742	632	690	701	826	2,573
15	83	90	77	81	13	
16					2,200	
17	903	823	829	862	2,232	
18	731	672	678	737	3,037	
19	91	87	73	85	2,317	
20					815	
21					4	

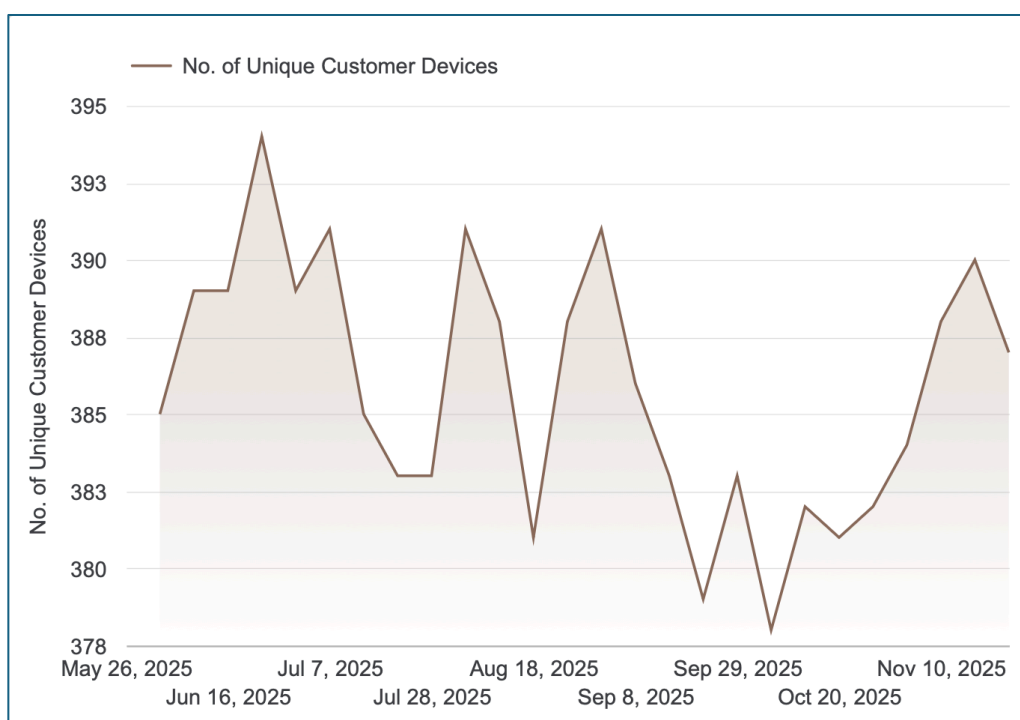
The heatmap shows unique customer devices by hour × day of week, highlighting clear peak-traffic windows. Activity is concentrated in late morning through early afternoon, with the strongest intensity on Thursday and Friday, where counts reach the highest values in the grid (e.g., Thursday/Friday midday levels in the ~3,000–4,000 range). This view supports practical decisions such as adjusting staffing coverage and service capacity during peak periods.

Figure 5.2 - Average Staffing per Shift by Role (Distinct Devices) (Query 5.2)



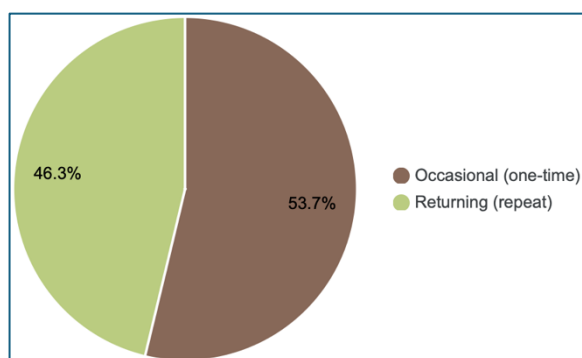
Average staffing (distinct employee devices) varies by shift and role. **Morning (06–13)** is the highest-staffed shift, with a broader role mix (cashier, general staff, delivery, butcher, security, senior staff, manager). **Afternoon (14–17)** remains strong but slightly leaner (delivery drops out), while **Evening (18–23)** is the leanest coverage overall. This supports aligning labour allocation to the demand peaks observed in the activity heatmap.

Figure 5.3 – Weekly Unique Customer Devices (*Query 5.3*)



Weekly unique customer devices (repeat + one-time) are relatively stable across the period, fluctuating within a narrow band (roughly 378–395 on the chart scale). This indicates consistent weekly reach, with only moderate week-to-week variation rather than a sustained upward/downward trend over Jun–Nov 2025.

Figure 5.4 - Customer Mix (Distinct Devices) (*Query 5.4*)



The customer mix by distinct device ID shows a slight skew toward Occasional (one-time) customers (53.7%) compared with Returning (repeat) customers (46.3%). For the selected Jun–Nov period, the total distinct customer devices shown in the dashboard is 761 (*Query 5.4*). This split helps frame retention opportunities (convert

occasional to returning) while tracking reach.